

SECTION B**9 (long)**

- (a) Find, with justification, the shortest distance between the lines
- r**
- and
- s**
- given by

$$\mathbf{r} = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} + u \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{s} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + v \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \quad [10]$$

- (b) Consider the following equation

$$(\mathbf{c} \cdot \mathbf{a})\mathbf{a} + (\mathbf{c} \cdot \mathbf{b})\mathbf{b} = \mathbf{p}$$

where **a** and **b** are known vectors which are not parallel.

- (i) What is the value of
- $\mathbf{p} \cdot (\mathbf{a} \times \mathbf{b})$
- ? [4]

- (ii) Explain carefully why
- c**
- can be written as

$$\mathbf{c} = \alpha \mathbf{a} + \beta \mathbf{b} + \gamma (\mathbf{a} \times \mathbf{b}) \quad [4]$$

- (iii) Given

$$\mathbf{a} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \quad \text{and} \quad \mathbf{p} = \begin{bmatrix} 2 \\ 0 \\ 8 \end{bmatrix}$$

find all solutions for **c**. [12]